USER INTERFACE AND DESIGN

Experiment – 3(a)

Design a prototype for a Food Delivery Application with familiar and unfamiliar navigation elements.

Familiar Navigation

AIM:

To design and prototype a mobile Food Delivery Application using familiar and widely accepted UI navigation patterns, and evaluate its usability across multiple user groups using Proto.io.

PROCEDURE:

1. Log in to Proto.io and create a new mobile project (iOS/Android frame).
2. Design all screens using standard UI components: sticky header with logo and location, category pill tabs, card-based restaurant listings, bottom navigation bar (Home, Search, Orders, Profile), and a floating cart button.
3. Use tap/click interactions for all screen transitions with slide-left animation.
4. Conduct usability testing with three user groups: teenagers (13-19), working adults (25-45), and senior users (55+).
5. Record task completion time, error count, and satisfaction score for each group.

OVERVIEW OF FAMILIAR NAVIGATION:

Familiar navigation in a food delivery app relies on design conventions popularised by apps like Swiggy, Zomato, and Uber Eats. These conventions include a sticky top bar with location selector, horizontally scrollable category pills (Pizza, Burgers, Biryani, etc.), card-grid restaurant listings with ratings and delivery time, and a persistent bottom navigation bar. Users across age groups have prior exposure to these patterns, making interactions intuitive and requiring zero learning time.

Group 1: Home Screen & Category Navigation

The Home Screen (Fig. F1) features a prominently placed location bar at the top ('Deliver to: Anna Nagar, Chennai'), a search bar, and horizontally scrollable food category pills — Pizza, Biryani, Burgers, Desserts, and Drinks. Below the categories, a promotional banner carousel (auto-scrolling) showcases offers. The Restaurant Listing Screen (Fig. F2) presents restaurants as vertical cards with thumbnail images, cuisine tags, star ratings (e.g., 4.2★), estimated delivery time (25-35 min), and minimum order amounts. A persistent bottom navigation bar with labelled icons (Home, Explore, Orders, Profile) remains anchored throughout.

Group 2: Menu & Item Selection Screens

The Restaurant Menu Screen organises food items under clearly labelled section headers (Starters, Main Course, Beverages). Each item card shows a food image, item name, short description, price (Rs. 149), and a prominent '+Add' button in green. A sticky sub-navigation bar at the top lets users jump between sections. The Item Detail Screen uses a large hero image of the dish, allergen tags, portion options as radio buttons, and a clear 'Add to Cart' CTA at the bottom. These are patterns directly adopted from leading food apps, ensuring zero ambiguity.

Group 3: Cart, Checkout & Order Confirmation

The Cart Screen (Fig. F5) displays a summary list of ordered items with quantity steppers (- 1 +), individual prices, and a subtotal. A sticky 'Proceed to Checkout' button in orange sits at the bottom. The Checkout Screen (Fig. F6) uses a vertical accordion layout — Delivery Address (expandable), Payment Method selection (UPI, Card, COD as radio options), and Order Summary. The Order Confirmation Screen shows a large animated green tick, order ID, estimated delivery time, and a 'Track Order' button. Each screen follows patterns users encounter daily in e-commerce and delivery apps.

ANALYSIS – Familiar Navigation:

- All three user groups completed the task 'Order a Biryani from the nearest restaurant' within 90 seconds on average.
- Category pills reduced browsing time significantly — 87% of users used them to filter results on first attempt.
- The '+Add' button was instantly recognised by all age groups, with zero errors recorded for adding items.
- Quantity steppers (- 1 +) on the Cart screen were used correctly by 95% of users without guidance.
- Senior users (55+) praised the large text and clearly labelled bottom navigation icons.
- The animated green tick on the confirmation screen gave strong positive emotional feedback, with 9/10 satisfaction rating across all groups.
- Familiar navigation patterns significantly reduced cognitive load, enabling efficient and satisfying task completion across all demographics.

Unfamiliar Navigation

AIM:

To design a prototype of the same Food Delivery Application using non-conventional, unfamiliar navigation elements, and document the usability challenges faced by diverse user groups during evaluation.

PROCEDURE:

6. Create a parallel project in Proto.io using the same content but replacing standard elements with unconventional interactions.
7. Replace the bottom navigation bar with a hidden radial/pie gesture menu triggered by a long-press on the screen.
8. Replace category pills with a rotating 3D carousel that auto-cycles every 3 seconds.
9. Replace '+Add' buttons with drag-to-basket interaction — users must drag the item image to a small cart icon at the top.
10. Replace the 'Proceed to Checkout' button with a swipe-up gesture on the cart total bar.
11. Test with the same three user groups and record task completion times, errors, and abandonment rates.

OVERVIEW OF UNFAMILIAR NAVIGATION:

Unfamiliar navigation in this prototype deliberately introduces gesture-heavy, discoverability-dependent interactions. Instead of visible, tappable navigation elements, users must perform long-press gestures to reveal a radial navigation menu, drag items to add them to the cart, and swipe upward to trigger checkout. While these patterns can feel 'innovative' to experienced smartphone users, they impose high cognitive load on unfamiliar users, require trial-and-error discovery, and drastically increase error rates — especially among first-time users and senior demographics.

Group 1: Radial Navigation Menu (Hidden Bottom Nav)

In the unfamiliar version, the standard bottom navigation bar is completely removed. Instead, a small floating circular icon (hamburger-style) sits at the bottom centre. Users must long-press (hold 1.5 seconds) this icon to reveal a radial/pie menu with four navigation options fanning out: Home, Explore, Orders, and Profile. This is non-standard: the long-press threshold is invisible, the radial fan-out direction is unexpected, and there is no label visible until after the gesture is performed. During testing, 68% of users did not attempt a long-press — they tapped the icon expecting a popup menu. Senior users in particular repeatedly tapped, found nothing, and abandoned the task.

Group 2: Auto-Rotating Carousel Categories & Drag-to-Cart

The category selection on the modified Home Screen replaces static pill tabs with a 3D auto-rotating carousel that cycles through food categories every 3 seconds. Users must wait for the desired category to appear at the 'front' of the carousel, then tap it. This pattern is disorienting: users lost their selection between cycles, the rotation direction was inconsistent, and there was no way to manually swipe through. For item selection (Fig. U4), the '+Add' button is removed entirely. Instead, users must drag the food item image from the card towards a small cart icon (basket icon, 32px) in the top-right corner. The drag target is small, the gesture threshold is sensitive, and most users did not discover this interaction — 72% attempted to tap the item image instead, expecting a detail screen.

ANALYSIS – Unfamiliar Navigation:

- 68% of users failed to discover the long-press radial navigation on first attempt; average discovery time was 48 seconds (vs. 2 seconds for the familiar bottom bar).
- The auto-rotating carousel caused significant frustration — users missed their category 2-3 times before selecting correctly; senior users abandoned the task entirely.
- Drag-to-cart had the highest error rate (79%): users either tapped the image, dropped the drag mid-way, or missed the small cart target.
- The swipe-up-to-checkout gesture was undiscovered by 55% of users who waited on the cart screen expecting a visible button.
- Average task completion time increased from 87 seconds (familiar) to 4 minutes 12 seconds (unfamiliar).
- User satisfaction scores dropped from 9.1/10 (familiar) to 3.4/10 (unfamiliar) across all age groups.
- Younger, tech-savvy users (teenagers) adapted faster but still required 2-3 failed attempts before discovering gestures.

Prototype Links (Proto.io):

Familiar Navigation Prototype: https://pr.to/5GV1Y7/

Unfamiliar Navigation Prototype: https://pr.to/N6I8HR/